Case 2: Operational Performance Benchmarking and Process Optimization

For a manufacturing client, our consulting team was tasked with benchmarking plant performance against industry standards and developing an optimisation roadmap to improve operational metrics like cycle time, downtimes, wastages, and costs.

Data Collection Template

As a consultant, you would start by designing a standardised Excel-based data collection template. This contained structured worksheets to capture the following:

Master data	Plant details, product/SKU data, machine information, etc.
Time study data	Detailed breakdowns of cycle times across processes.
Downtime logs	Instances and reasons for process disruptions.
Rejection data	Product rejections/wastages and root causes
Manpower logs	Operator allocation, efficiencies, etc.
KPI tracking	Computations for metrics like OEE, MTTR, yield rates, etc

Data Validation & Interactive Dashboards

As various teams share plant data, you can implement several data validation rules and checks in the template using the following:

- Dropdowns with permissible options for error-free master data entry
- Conditional formatting to flag incorrect/missing data in time study logs
- SUMPRODUCT formulas to cross-check the matching of process times across sources
- COUNTIFS criteria to identify outliers in downtime/rejection reasons